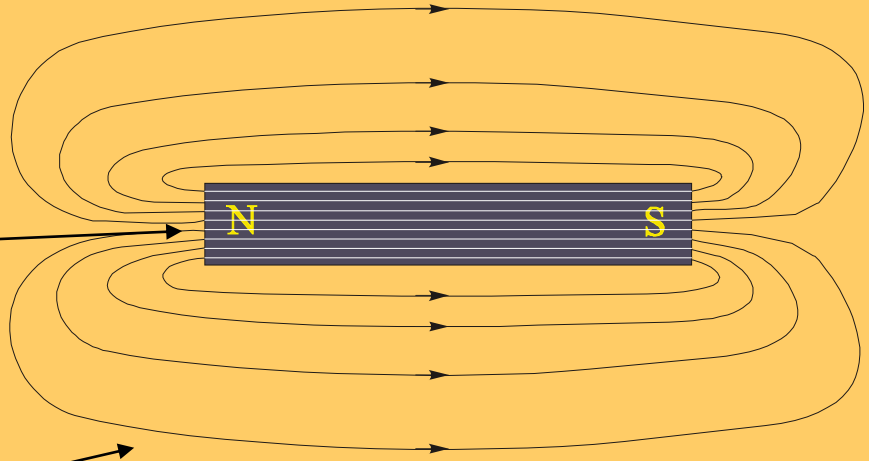
MAGNETIC QUANTITIES & MAGNETIZATION CURVE

Magnetic Quantities

Magnetic fields are described by drawing flux lines that represent the magnetic field.

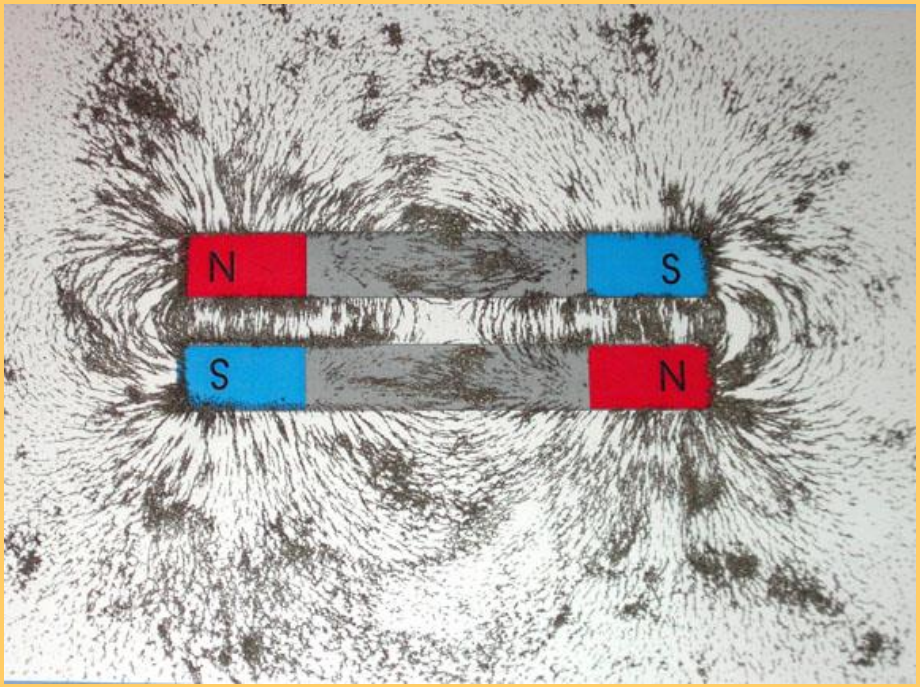
Where lines are close together, the flux density is higher.



Where lines are further apart, the flux density is lower.

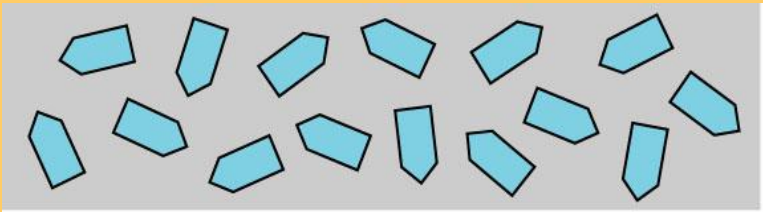
Magnetic Quantities

Magnetic flux lines are invisible, but the effects can be visualized with iron filings sprinkled in a magnetic field.

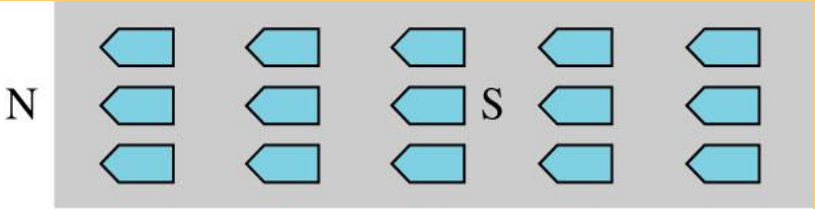


Magnetic Materials

Ferromagnetic materials such as iron, nickel and cobalt have randomly oriented magnetic domains, which become aligned when placed in a magnetic field, thus they effectively become magnets.



(a) The magnetic domains (N S) are randomly oriented in the unmagnetized material.



(b) The magnetic domains become aligned when the material is magnetized.

Magnetic Quantities

The unit of flux is the weber. The unit of flux density is the weber/square meter, which defines the unit **tesla**, (T), which is a very large unit.

Flux density is given by the equation

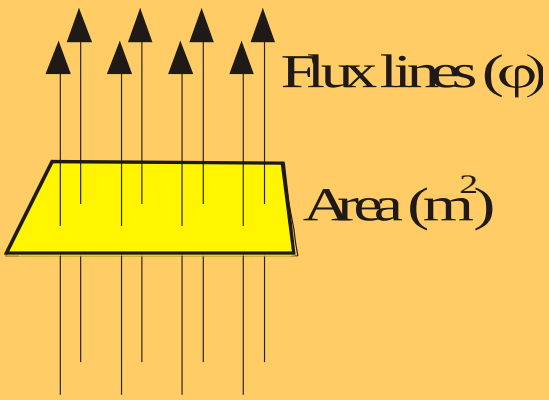
$$B = \frac{\varphi}{A}$$

where

B = flux density (T)

φ = flux (Wb)

A = area (m^2)



Magnetic Quantities

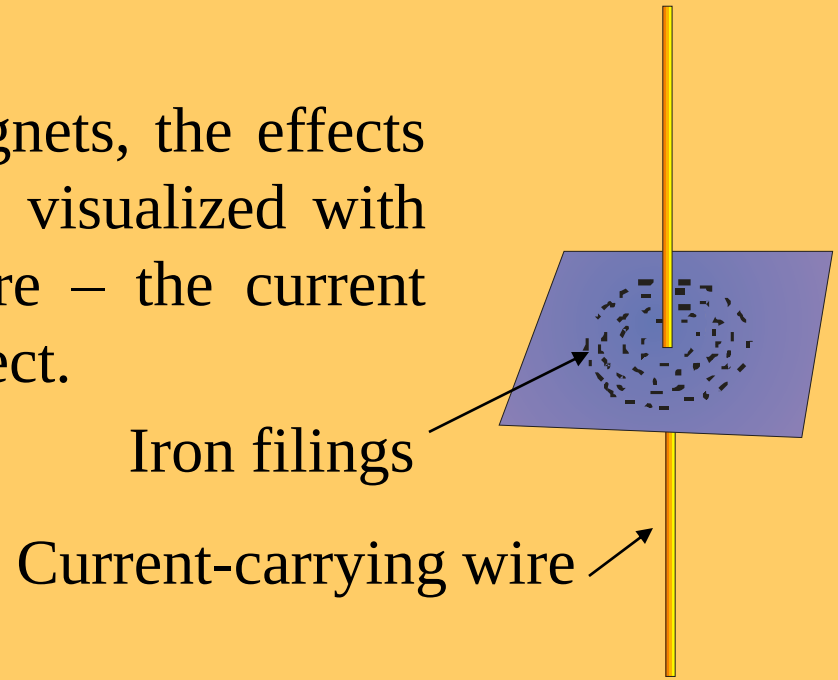
Example: What is the flux density in a rectangular core that is 8 mm by 10 mm if the flux is 4 mWb?

$$B = \frac{\varphi}{A}$$

$$B = \frac{4 \times 10^{-3}\ \text{Wb}}{\left(8 \times 10^{-3}\ \text{m}\right)\left(10 \times 10^{-3}\ \text{m}\right)} = 50\ \text{Wb/m}^2 = 50\ \text{T}$$

Magnetic Quantities

- Magnetic flux lines surround a current carrying wire.
- The field lines are concentric circles as shown in Figure 10.
- As in the case of bar magnets, the effects of electrical current can be visualized with iron filings around the wire – the current must be large to see this effect.



Magnetic Quantities

- **Permeability** (μ) defines the ease with which a magnetic field can be established in a given material. It is measured in units of the weber per ampere-turn meter.

- The permeability of a vacuum (μ_0) is $4\pi \times 10^{-7}$ weber per ampere-turn meter, which is used as a reference.

- **Relative Permeability** (μ_r) is the ratio of the absolute permeability to the permeability of a vacuum.

$$\mu_r = \frac{\mu}{\mu_0}$$

Magnetic Quantities

• **Reluctance** ($\mathcal{R}$) is the opposition to the establishment of a magnetic field in a material.

$$\mathcal{R} = \frac{l}{\mu A}$$

$\mathcal{R}$= reluctance in A-t/Wb
l = length of the path
μ = permeability (Wb/A-t m).
A = area in m^2

Magnetic Quantities

- Recall that magnetic flux lines surround a current-carrying wire. A coil reinforces and intensifies these flux lines.

- The *cause* of magnetic flux is called magnetomotive force (mmf), which is related to the current and number of turns of the coil.

$$F_m = NI$$

F_m = magnetomotive force (A-t)
N = number of turns of wire in a coil
I = current (A)

Magnetic Quantities

- Ohm's law for magnetic circuits is

$$\varphi = \frac{F_m}{\mathcal{R}}$$

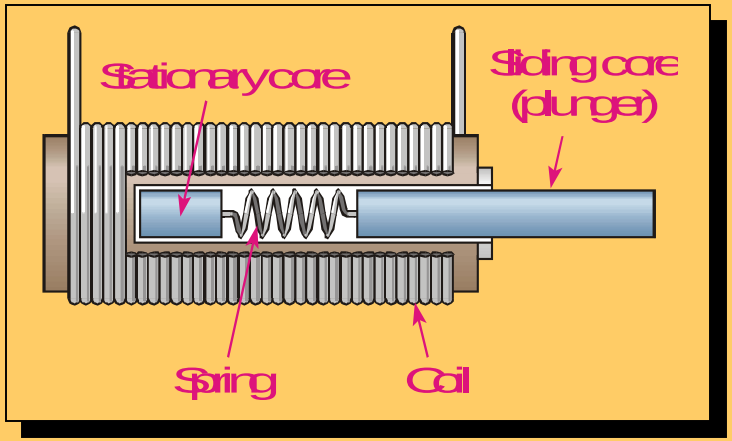
- flux (φ) is analogous to current
- magnetomotive force (F_m) is analogous to voltage
- reluctance ($\mathcal{R}$) is analogous to resistance.

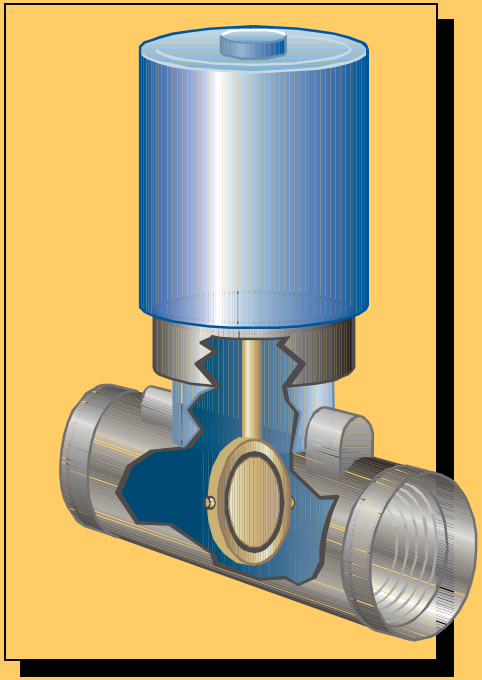
Problem: What flux is in a core that is wrapped with a 300 turn coil with a current of 100 mA if the reluctance of the core is 1.5×10^7 A-t/Wb ?

2.0 μWb

Solenoids

•A **solenoid** produces mechanical motion from an electrical signal.





One application is valves that can remotely control a fluid in a pipe, such as in sprinkler systems.

• **Magnetic field intensity** is the magnetomotive force per unit length of a magnetic path.

$$H = \frac{F_m}{l} \quad \text{or} \quad H = \frac{NI}{l}$$

H = Magnetic field intensity (Wb/A-t m)
F_m = magnetomotive force (A-t)
l = average length of the path (m)
N = number of turns
I = current (A)

• Magnetic field intensity represents the effort that a given current must put into establishing a certain flux density in a material.

Magnetic Quantities

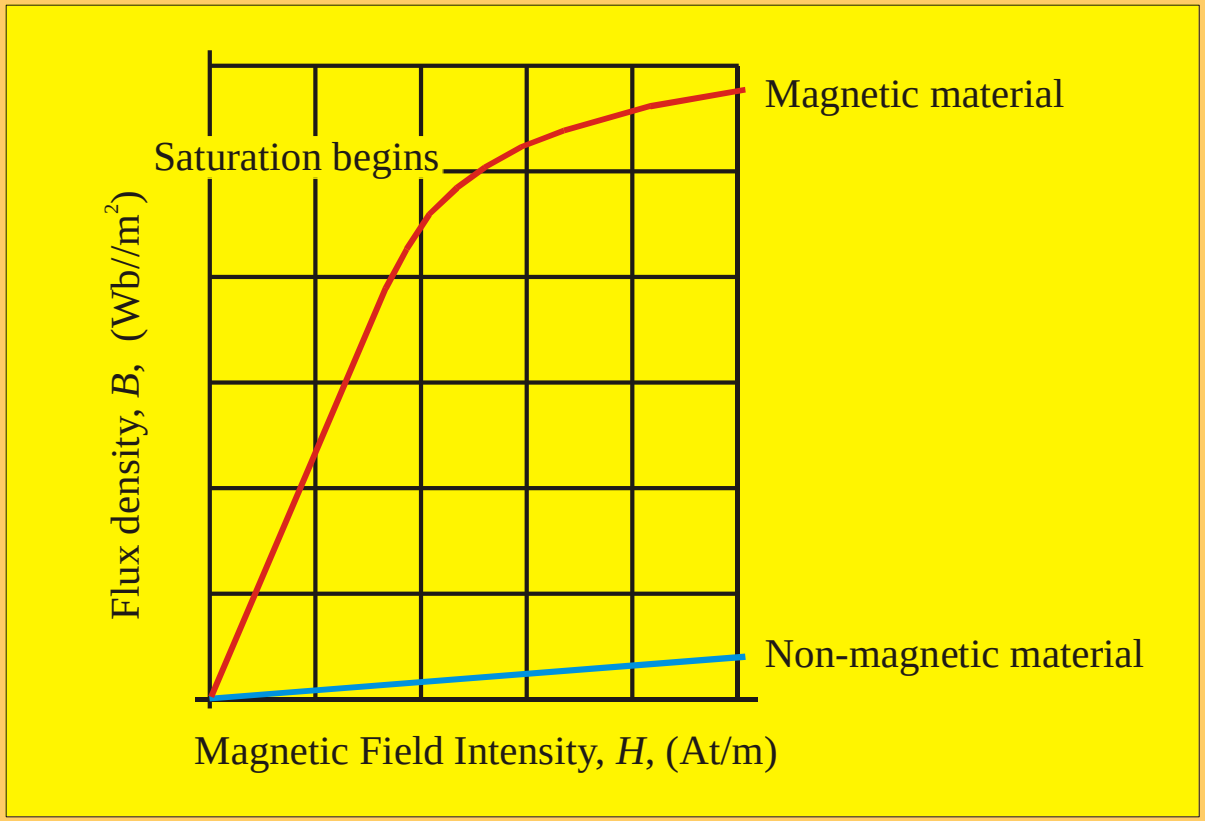
- If a material is permeable, then a greater flux density will occur for a given magnetic field intensity. The relation between B (flux density) and H (the effort to establish the field) is

$$\boxed{B = \mu H}$$

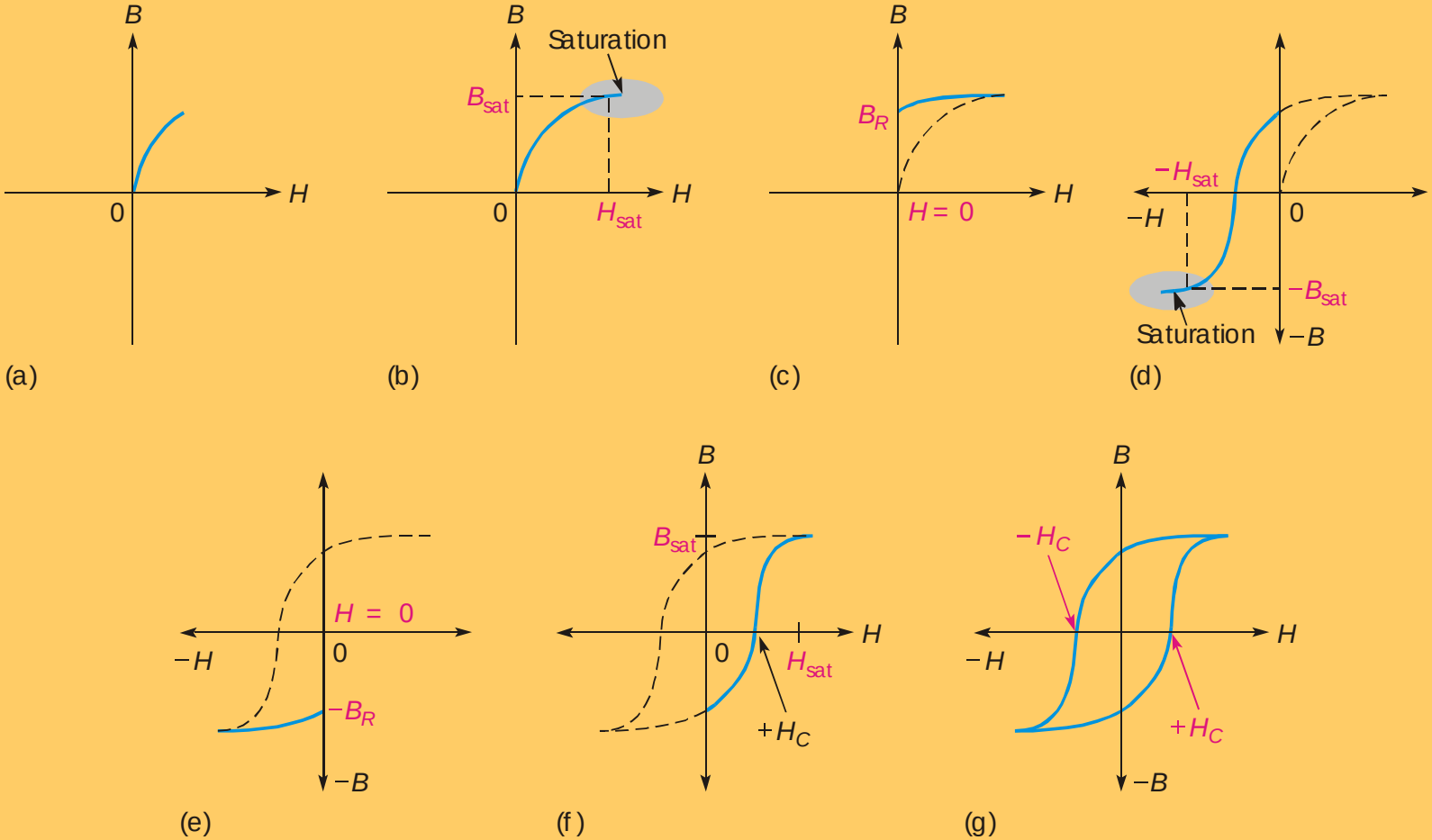
μ = permeability (Wb/A-t m).
H= Magnetic field intensity (Wb/A-t m)

- This relation between B and H is valid up to saturation, when further increase in H has no affect on B.

• As the graph shows, the flux density depends on both the material and the magnetic field intensity.



As H is varied, the magnetic hysteresis curve is developed.

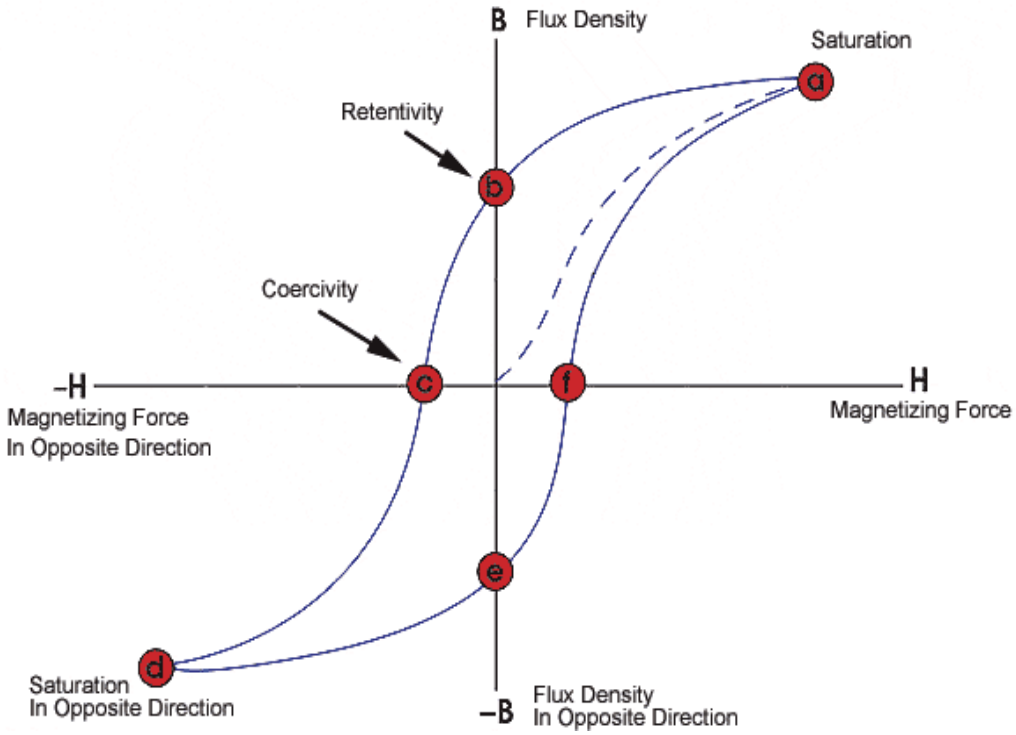


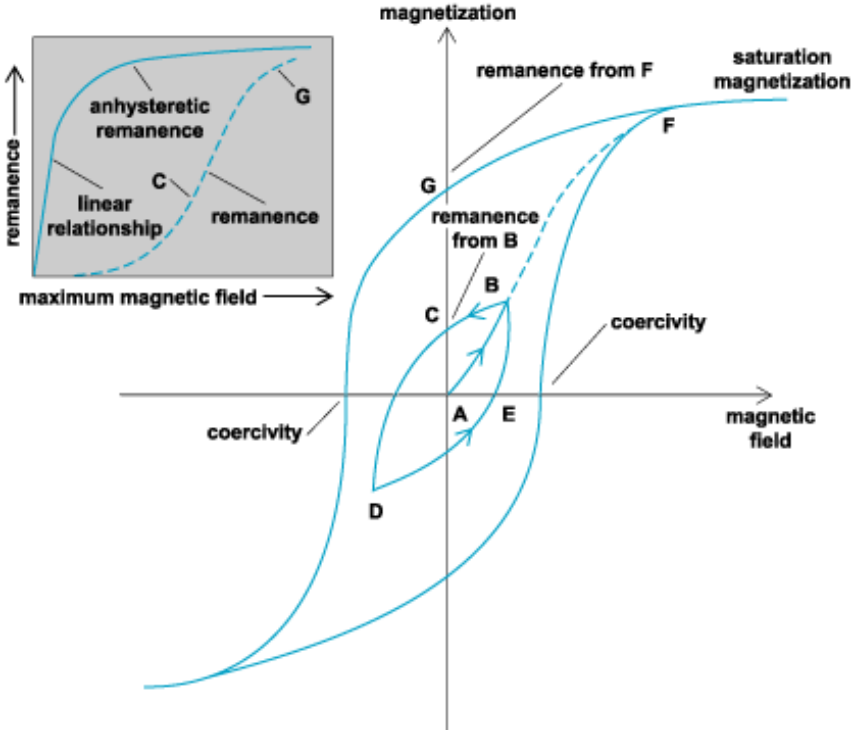
Magnetization Curve

Magnetization Curve: A graph of magnetic flux density against magnetic field strength in the same region *i.e.* a graph representing changes in the condition of a magnetizable substance with magnetizing force ***H*** as abscissa and either magnetization ***I*** or induction ***B*** as ordinate.

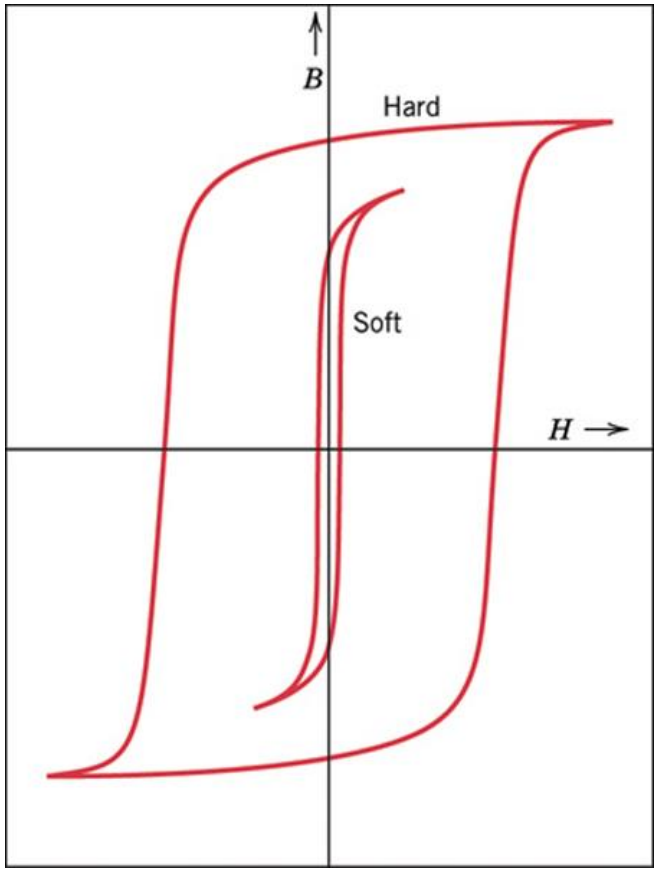
Magnetization Curve: A characteristic curve of an electromagnet, indicating the relation of magnetization to exciting current. Laying off on the axis of ordinates the quantities of magnetism evoked, and the corresponding strengths of the exciting current on the axis of abscissas, the curve can be plotted. It first rises rapidly, indicating a rapid increase of magnetization, but grows nearly horizontal as the iron becomes more saturated. The effect due to the coils alone, or the effect produced in the absence of iron is a straight line, because air does not change in permeability.

Magnetization Curve





Hard vs. Soft Magnets



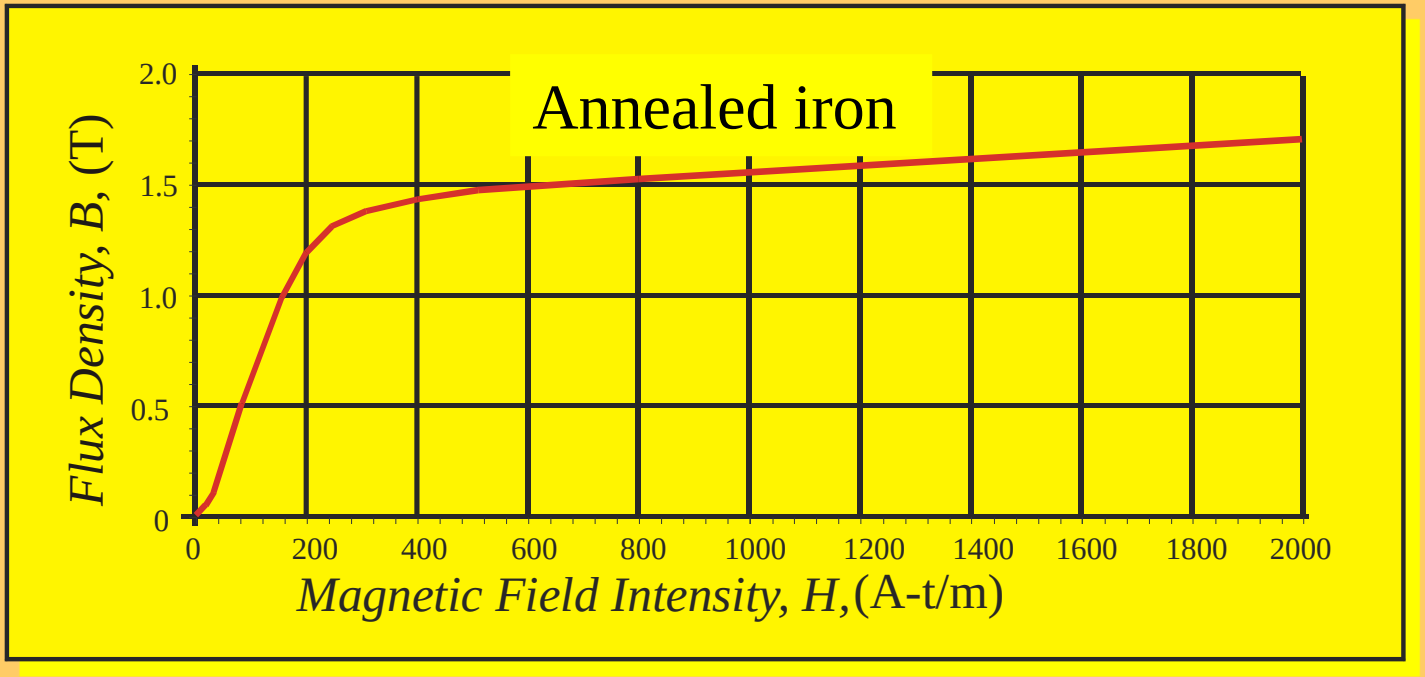
Hard magnetic materials retain magnetism after field is removed; **soft** magnetic materials do not.

Soft magnetic materials require small H to reach $\mathbf{B_s}$, and $\mathbf{H_c}$ is small. This means less energy is wasted in hysteresis loop. Soft magnetic materials are desirable in transformer cores and other applications where residual magnetization is undesirable

Defects, such as nonmagnetic phases and voids restrict easy movement of domain walls and are to be avoided in soft magnetic materials. Sometimes Si or Ni is added to Fe to minimize *eddy currents* that rob energy in hysteresis loops

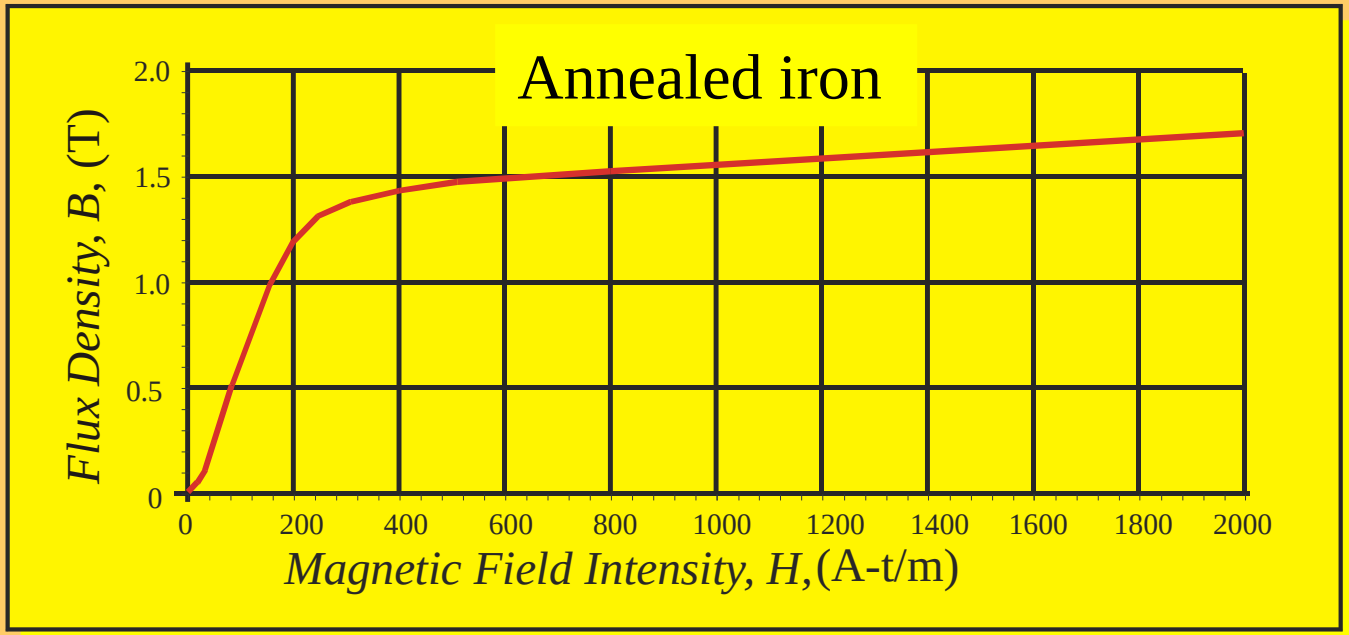
Magnetization Curve

A *B-H* curve is referred to as a **magnetization curve** for the case where the material is initially unmagnetized.



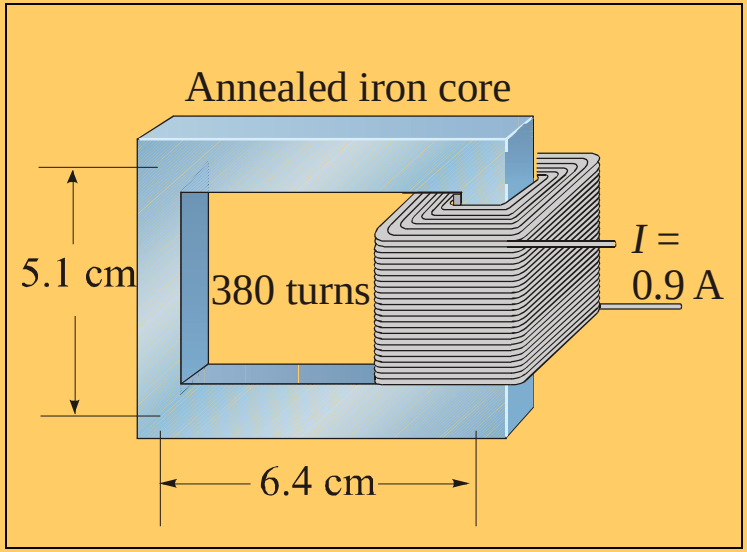
Magnetization Curve

A *B-H* curve can be read to determine the flux density in a given core. The next slide shows how to read the graph to determine the flux density in an annealed iron core.



Magnetization Curve

Example: What is B for the core?

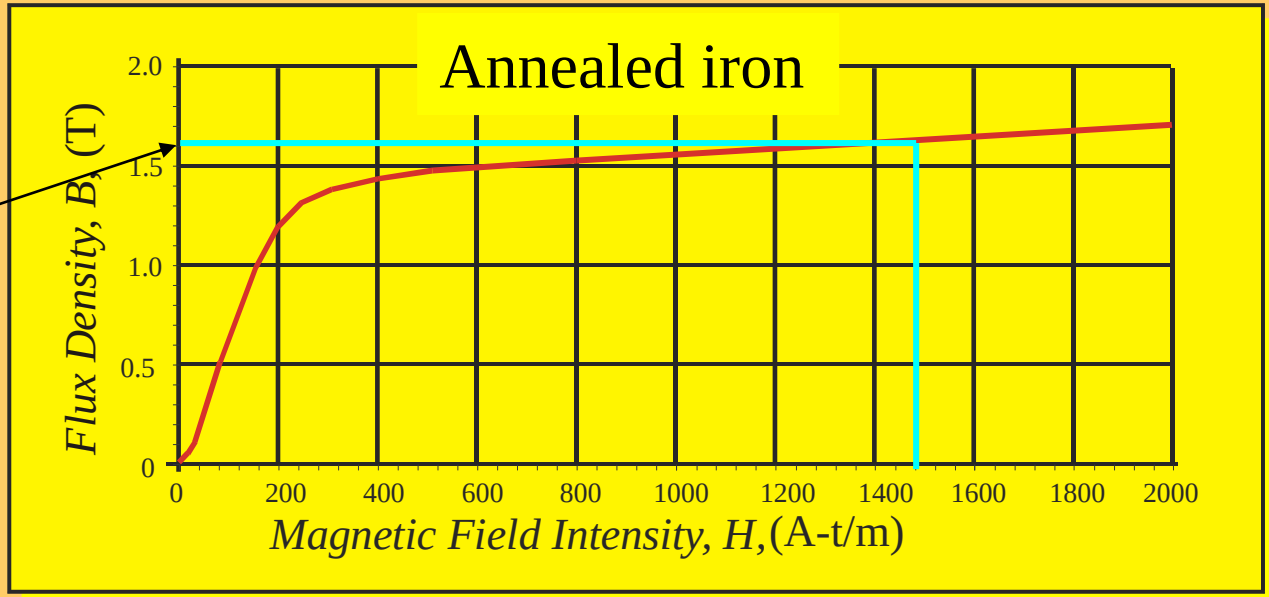


Solution:

$$H = \frac{NI}{l} = \frac{(380\text{ t})(0.9\text{ A})}{0.23\text{ m}} = 1487\text{ A-t/m}$$

Reading the graph,

B = 1.63 T



l=5.1*2+6.4*2=23 cm

Selected Key Terms

Magnetic field	A force field radiating from the north pole to the south pole of a magnet.
Magnetic flux	The lines of force between the north pole and south pole of a permanent magnet or an electromagnet.
Weber (Wb)	The SI unit of magnetic flux, which represents 10^8 lines.
Permeability	The measure of ease with which a magnetic field can be established in a material.
Reluctance	The opposition to the establishment of a magnetic field in a material.

Magnetic units

It is useful to review the key magnetic units from this chapter:

Quantity	SI Unit	Symbol
Magnetic flux density	Tesla	B
Flux	Weber	ϕ
Permeability	Weber/ampere-turn-meter	μ
Reluctance	Ampere-turn/Weber	$\mathcal{R}$
Magnetomotive force	Ampere-turn	F_m
Magnetizing force	Ampere-turn/meter	H

Selected Key Terms

Magnetomotive force (mmf)	The cause of a magnetic field, measured in ampere-turns.
Solenoid	An electromagnetically controlled device in which the mechanical movement of a shaft or plunger is activated by a magnetizing current.
Hysteresis	A characteristic of a magnetic material whereby a change in magnetism lags the application of the magnetic field intensity.
Retentivity	The ability of a material, once magnetized, to maintain a magnetized state without the presence of a magnetizing current.

Selected Key Terms

Induced voltage (v_{ind})	Voltage produced as a result of a changing magnetic field.
Faraday's law	A law stating that the voltage induced across a coil of wire equals the number of turns in the coil times the rate of change of the magnetic flux.
Lenz's law	A law stating that when the current through a coil changes, the polarity of the induced voltage created by the changing magnetic field is such that it always opposes the change in the current that caused it. The current cannot change instantaneously.